

1 ZebraPuzzle Example

Setup

 Houses	1, 2, 3
 Person	Alice, Bob, Carol
 Pet	Cat, Dog, Fish
 Drink	Tea, Coffee, Juice

Original Natural-Language Clues

- C1. Alice lives immediately to the left of the cat owner.
- C2. Bob does not drink coffee.
- C3. House 2 serves tea.

2 Parsed Syntactic Clues

Clues converted to solver-checkable form

- C1: Alice + 1 == Cat.
- C2: Bob != Coffee.
- C3: Tea == 2.
- Alice, Bob, Carol → 1, 2, 3
- Cat, Dog, Fish → Cat, Dog, Fish
- Tea, Coffee, Juice → Tea, Coffee, Juice

Notation: + 1 means immediately; to the right; == means “is in”; != means “is not in”.

3 Interleaved Reasoning Steps

- R1

From C1, Alice cannot be in House 3.
- S1

Alice != 3.
- R2

From C3, House 2 serves tea.
- S2

Tea == 2.
- R3

Therefore House 2 cannot serve coffee.
- S3

Coffee != 2.
- R4

Bob drinks coffee.
- S4

Bob == Coffee.
- R5

Bob cannot be the tea drinker.
- S5

Bob != 2.

- R: Natural language reasoning
- S: Syntactic step (solver-checkable)

4 Solver Evaluation

Step	Status	Why
S1	Valid, Novel	Follows from left-of relation.
S2	Valid, but Clue Restatement	Directly from C3.
S3	Valid, Novel	Derived from Tea == 2.
S4	Contradictory	Conflicts with C2 (Bob does not drink coffee).
S5	Valid, Novel	Combines C2 and C3.

- Valid, Novel (syntactic + informative)
- Valid, but Clue Restatement (non-novel)
- Contradictory

5 Process Reward Signal

- + reward for Valid + Novel steps
- 0 or small reward for Valid but non-novel steps
- penalty for Contradictory steps

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 Valid steps: 4
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 Novel steps: 3
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 Contradictions: 1

Final process reward: Positive

Current solution (illustrative)

House	Person	Drink	Pet
1	Alice	–	–
2	–	Tea	–
3	Bob	Coffee	–

Reasoning trajectory is shaped using solver feedback.